22: Final Project

Engr 315: Hardware / Software Codesign

Andrew Lukefahr
Indiana University

P6 Dot

- Slight changes to Project.

Starts @ 2210

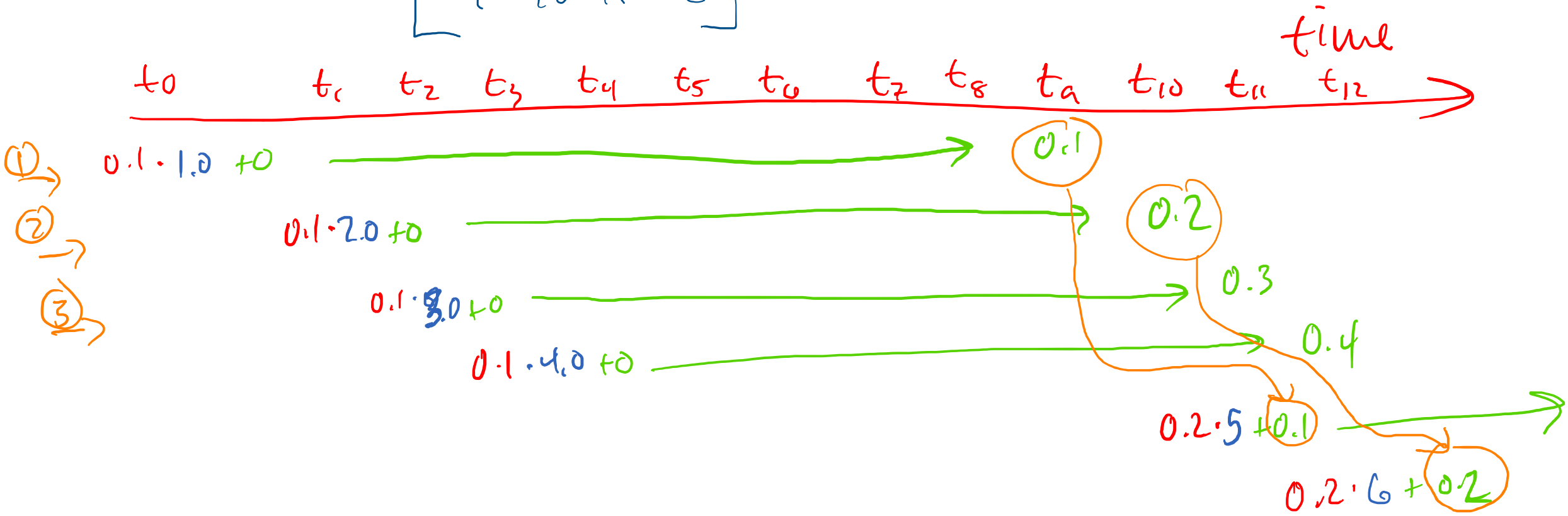
- < 500 cycles for a 60% score on the Autograder
- < 300 cycles for a 100% score on the Autograder
- < 200 cycles for a 105% score on the Autograder

P6 Tips ‘n’ Tricks

- Groups of 2?
- Pipelining:
  - ONLY update state machine in dot.sv.
- Parallelizing:
  - Not recommended: multiple fmac’s in dot.sv
  - Recommended: multiple dot.sv’s in dot_20_10.sv

Pipelining

$$\begin{bmatrix} 0.1 & 0.2 & 0.3 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{bmatrix} = \begin{bmatrix} 3.8 & 4.4 & 5 & 5.6 \end{bmatrix}$$



Dot Pipelining

$$\begin{bmatrix} 0.1 & 0.2 & 0.3 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{bmatrix} = \begin{bmatrix} 3.8 & 4.4 & 5 & 5.6 \end{bmatrix}$$

1. $0.1 \cdot 1.0 + 0 = 0.1$
2. $0.1 \cdot 2.0 + 0 = 0.2$
3. $0.1 \cdot 3.0 + 0 = 0.3$
4. $0.1 \cdot 4.0 + 0 = 0.4$

(input

P6 Pipelining

$$\begin{bmatrix} 0.1 & 0.2 & 0.3 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{bmatrix} = \begin{bmatrix} 3.8 & 4.4 & 5 & 5.6 \end{bmatrix}$$

$0.1 \cdot 1 + 0 = 0.1$

$0.1 \cdot 2 + 0 = 0.2$

$0.1 \cdot 3 + 0 = 0.3$

$0.1 \cdot 4 + 0 = 0.4$

$0.2 \cdot 5 + 0.1 = 1.1$

$0.2 \cdot 6 + 0.2 = 1.4$

$0.2 \cdot 7 + 0.3 = 1.7$

$0.2 \cdot 8 + 0.4 = 2.0$

$0.3 \cdot 9 + 1.1$

$0.3 \cdot 10 + 1.4$

$0.3 \cdot 11 + 1.7$

$0.3 \cdot 12 + 2.0$

$= 3.8$

4.4

5.0

5.6

P6 Demos

P6 Parallelization

- Option 1: Multiple FMACs
- Option 2: Multiple Dots, Weights split vertically
- Option 3: Multiple Dots, weights split horizontally

Option 1: Multiple FMACs

$$[0.1 \quad 0.2 \quad 0.3 \quad 0.4] \cdot \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix} = [9 \quad 10 \quad 11 \quad 12]$$

time

t_0 t_1 t_2 t_3

$0.1 \cdot 1 + 0$

$0.1 \cdot 2 + 0$

$0.1 \cdot 3 + 0$

$0.1 \cdot 4 + 0$

Option 2: Multiple Dots, Weights Split Vertically

$$\begin{bmatrix} 0.1 & 0.2 & 0.3 & 0.4 \end{bmatrix} \cdot \left[\begin{array}{cc|cc} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{array}\right] = \begin{bmatrix} 9 & 10 & 11 & 12 \end{bmatrix}$$

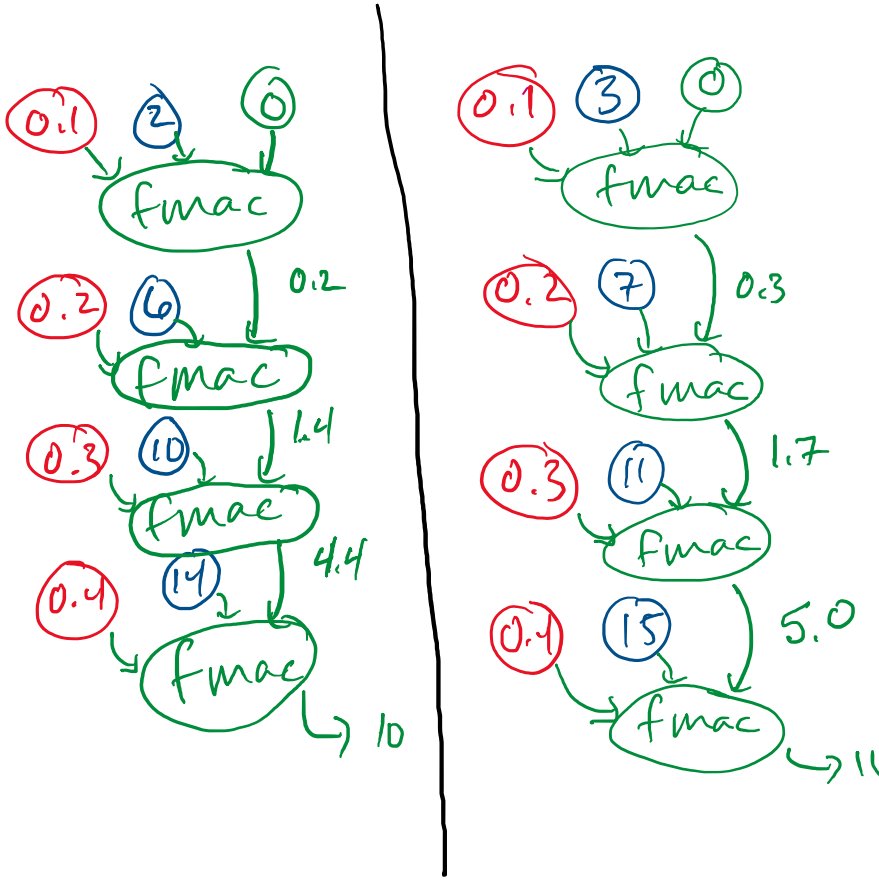
$$\begin{bmatrix} 0.1 & 0.2 & 0.3 & 0.4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 5 & 6 \\ 9 & 10 \\ 13 & 14 \end{bmatrix} = \begin{bmatrix} 9 & 10 \end{bmatrix}$$

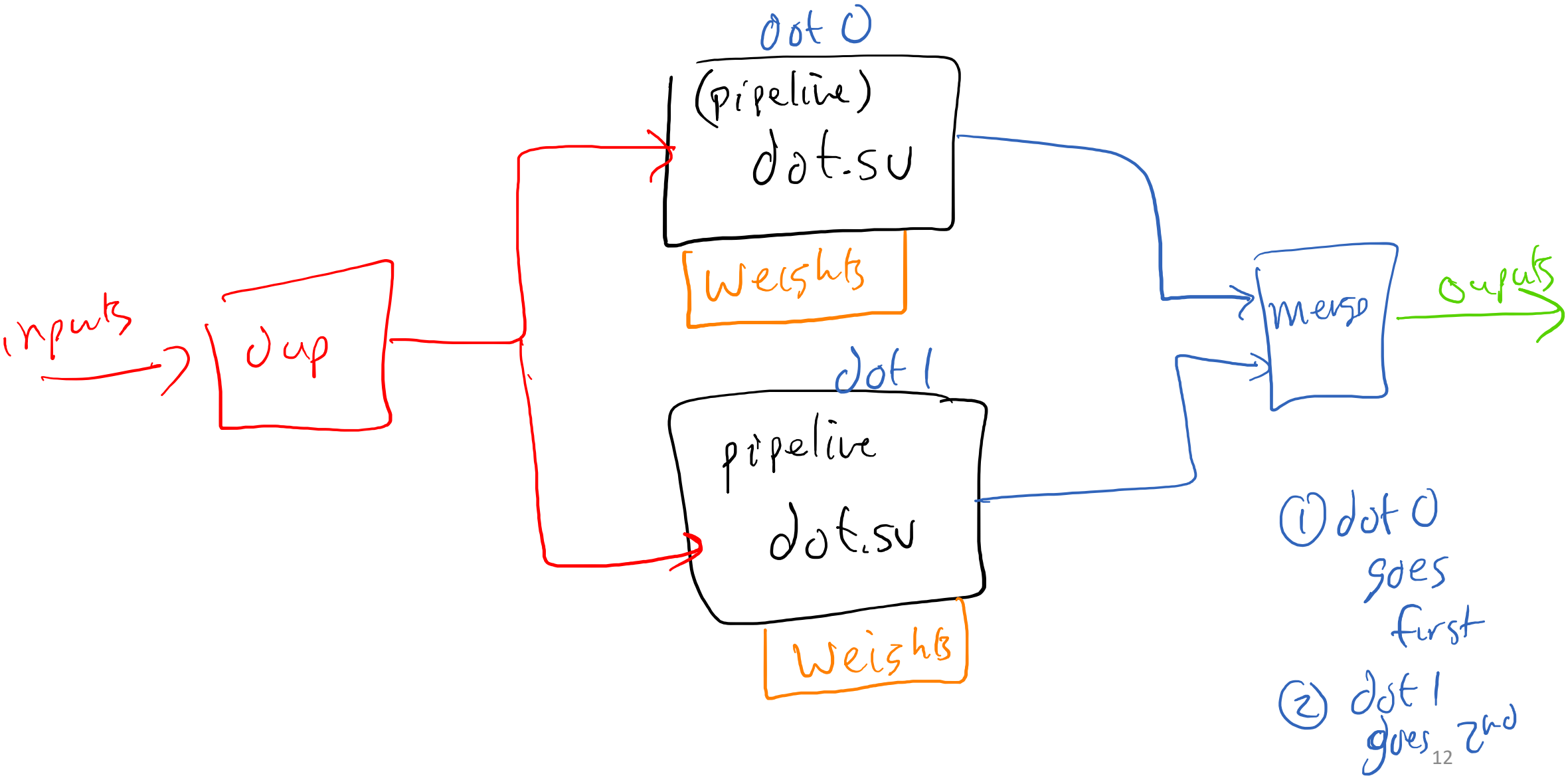
$$\begin{bmatrix} 0.1 & 0.2 & 0.3 & 0.4 \end{bmatrix} \begin{bmatrix} 3 & 4 \\ 7 & 8 \\ 11 & 12 \\ 15 & 16 \end{bmatrix} \begin{bmatrix} 11 & 12 \end{bmatrix}$$

$$\begin{bmatrix} 0.1 & 0.2 & 0.3 & 0.4 \end{bmatrix} \cdot \left[\begin{array}{cc|cc} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{array}\right] = \begin{bmatrix} 9 & 10 & 11 & 12 \end{bmatrix}$$

$$\begin{bmatrix} 0.1 & 0.2 & 0.3 & 0.4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 5 & 6 \\ 9 & 10 \\ 13 & 14 \end{bmatrix} = \begin{bmatrix} 9 & 10 \end{bmatrix}$$

$$\begin{bmatrix} 0.1 & 0.2 & 0.3 & 0.4 \end{bmatrix} \begin{bmatrix} 3 & 4 \\ 7 & 8 \\ 11 & 12 \\ 15 & 16 \end{bmatrix} = \begin{bmatrix} 11 & 12 \end{bmatrix}$$




dot 0
(pipeline)
dot.sv
weights
inputs
dup
dot 1
pipeline
dot.sv
weights
merge
outputs
① dot 0 goes first
② dot 1 goes 2nd

Option 2: Multiple Dots, Weights Split Horizontally

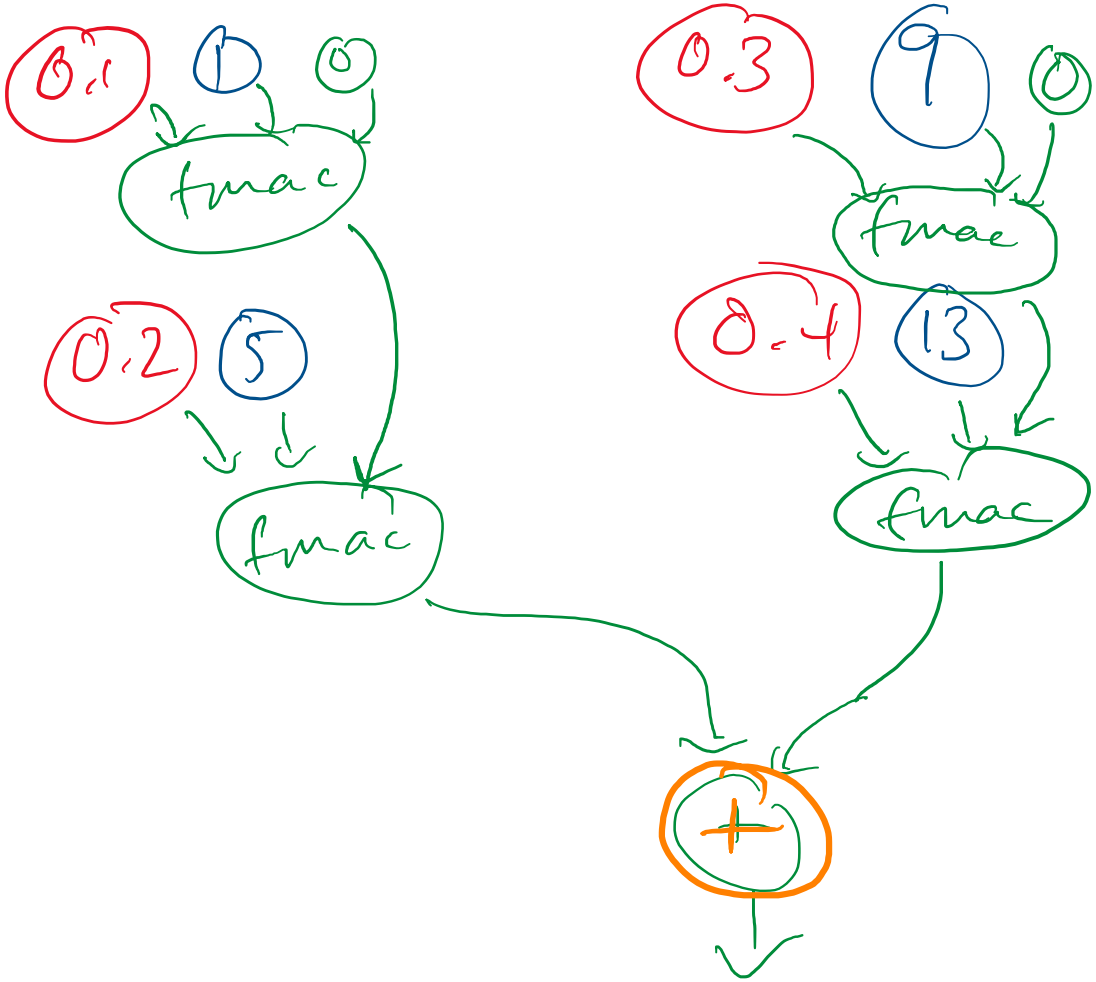
$$\begin{bmatrix} 0.1 & 0.2 & 0.3 & 0.4 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix} = \begin{bmatrix} 9 & 10 & 11 & 12 \end{bmatrix}$$

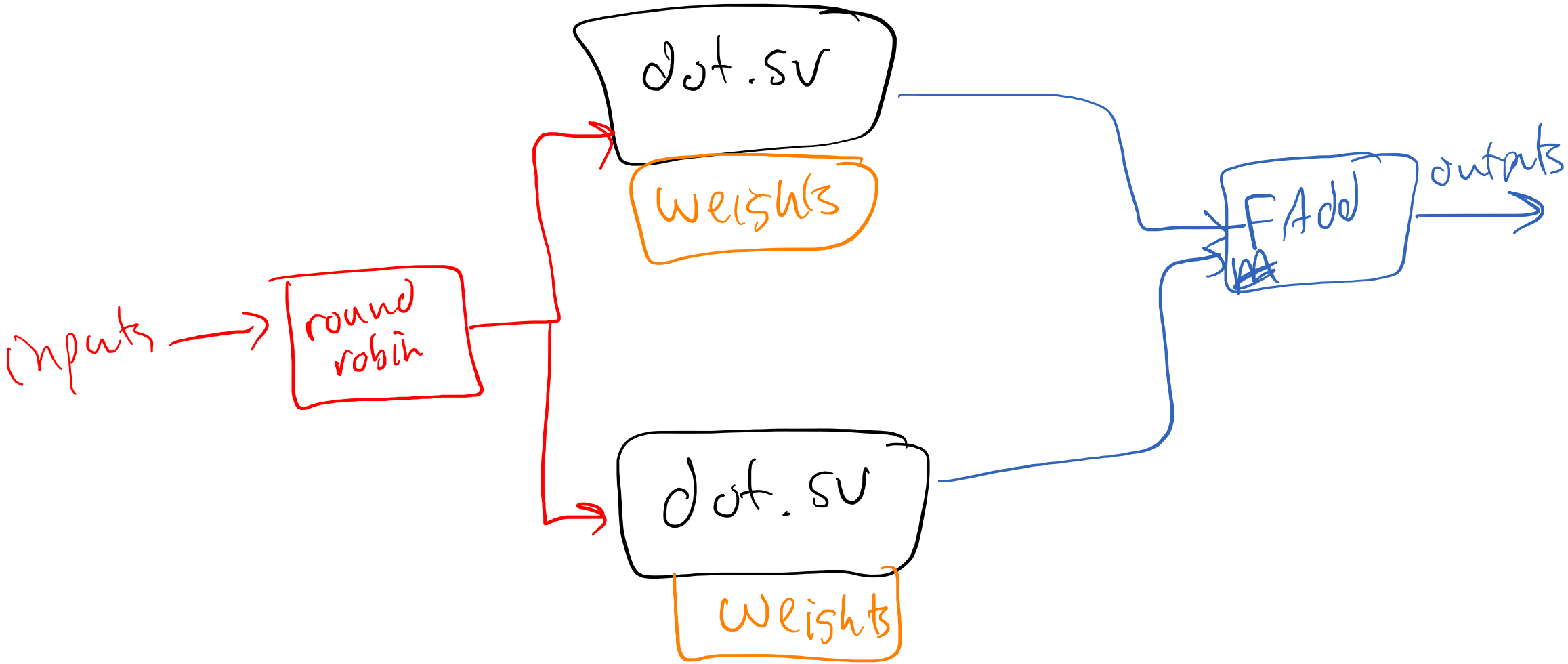
$$[0.1 \quad 0.2 \quad 0.3 \quad 0.4] \cdot \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ \hline 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix} = [9 \quad 10 \quad 11 \quad 12]$$

$$\rightarrow [0.1 \quad 0.2] \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix}$$

$$+ \; [0.3 \quad 0.4] \begin{bmatrix} 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix}$$

$$[9 \quad 10 \quad 11 \quad 12]$$




inputs
round robin
dot.sv
weights
dot.sv
weights
FAdd
outputs

Final Project

- Accelerate Dot Project
- Bigger Dot Product
  - 1st layer of MNIST digit recognition
- Drop the remaining MNIST stuff, not enough time.

Final project

- Any combination of CPUs, FPGA
- Score based on overall speedup
  - → 2x CPUs
  - → 4x FPGAs
  - → 10x CPUs + FPGA

Final Report

- What did you do?
- What worked, what didn't? ***WHY?***
- **What did you learn?**
- What would you do different in the future?

- **Report grade independent of project grade.**

- **Takes the place of the final.**

22: Final Project

Engr 315: Hardware / Software Codesign

Andrew Lukefahr
Indiana University